TECHVID INNOVATIONS

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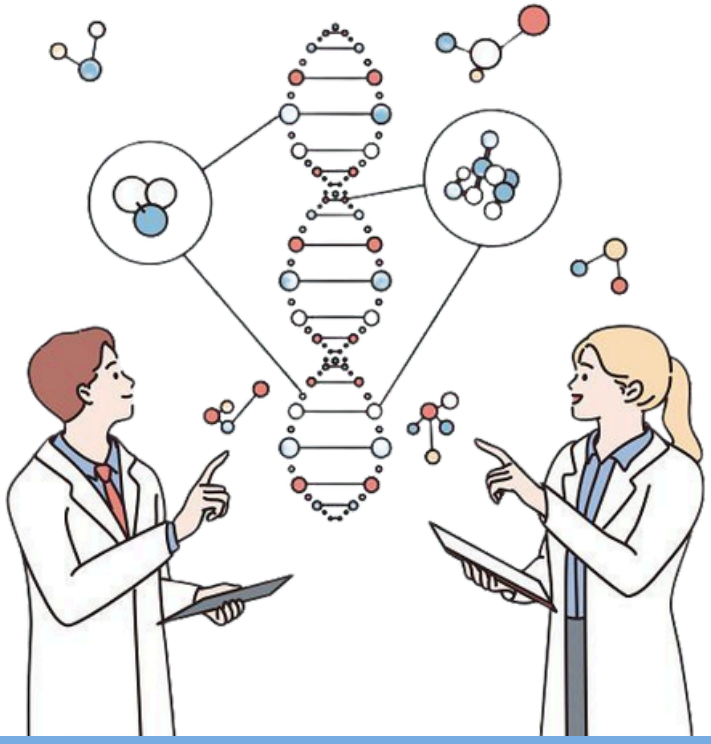
GENETIC ENGINEERING (AGRICULTURE)

From Beginner to Industry Ready



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Introduction to Genetic Engineering(Agriculture)

Genetic Engineering in agriculture involves modifying the genetic material of plants to improve traits such as yield, resistance to pests, diseases, and environmental conditions. It plays a crucial role in modern farming by enhancing crop productivity, ensuring food security, and supporting sustainable agricultural practices. This field combines biotechnology, molecular biology, and agriculture to develop improved crop varieties.

FUNDAMENTALS OF GENETIC ENGINEERING

Introduction to Genetic Engineering

What is Genetic Engineering | Importance in Agriculture | Applications | Advantages & Limitations

Basics Of Genetics

DNA Structure | Genes | Chromosomes | Gene Expression | Genetic Variation

Cell Biology

Plant Cell Structure | Cell Division | Tissue Culture | Cell Components

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MOLECULAR BIOLOGY TECHNIQUES

DNA & RNA Techniques

DNA Extraction | RNA Isolation | Gel Electrophoresis | PCR (Polymerase Chain Reaction)

Recombinant DNA Technology

Restriction Enzymes | DNA Cloning | Vectors | Gene Insertion

Gene Editing

CRISPR-Cas9 | Gene Silencing | Genome Editing Techniques

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PLANT BIOTECHNOLOGY

Plant Tissue Culture

Callus Culture | Micropropagation | Somatic Embryogenesis | Regeneration

Genetic Transformation

Agrobacterium-mediated Transformation | Gene Transfer Methods | Transgenic Plants

Crop Improvement

Disease Resistance | Pest Resistance | Drought Tolerance | Yield Enhancement

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BIOINFORMATICS & DATA ANALYSIS

Bioinformatics Basics

Sequence Analysis | Databases | Gene Mapping | Data Interpretation

Tools & Software

BLAST | GenBank | Genome Browsers | Data Analysis Tools

Statistical Analysis

Data Collection | Analysis Methods | Interpretation

REGULATIONS & ETHICS

Biosafety Regulations

GM Crop Regulations | Risk Assessment | Safety Guidelines

Ethical Issues

Environmental Impact | Food Safety | Public Concerns

Regulatory Bodies

ICAR | DBT | International Guidelines

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PROJECTS

Mini Projects

DNA Extraction Experiment
PCR Analysis
Plant Tissue Culture
Gene Analysis

Major Projects

Transgenic Plant Development
CRISPR-based Gene Editing
Crop Improvement Study
Bioinformatics Analysis Project

TOOLS COVERED

eCTD Software
Document Management Systems
Regulatory Databases
MS Excel
Clinical Trial Systems
Publishing Tools
Submission Portals
Documentation Tools